In [11]:
```
# ==============================
# Logistic Regression Example
# Buys_Computer Dataset
# ==============================

import pandas as pd
from sklearn.model_selection import train_test_split
#from sklearn.linear_model import LogisticRegression

from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, classification_report
```

In [12]:
```
dataset=pd.read_csv("/home/mvsr/Desktop/Customer.csv")
```

In [13]:
```
dataset
```

Out[13]:

	Age	Income	Student	Credit_Rating	Buys_Computer
0	youth	high	no	fair	no
1	youth	high	no	excellent	no
2	middle_aged	high	no	fair	yes
3	senior	medium	no	fair	yes
4	senior	low	yes	fair	yes
5	senior	low	yes	excellent	no
6	middle_aged	low	yes	excellent	yes
7	youth	medium	no	fair	no
8	youth	low	yes	fair	yes
9	senior	medium	yes	fair	yes
10	youth	medium	yes	excellent	yes
11	middle_aged	medium	no	excellent	yes
12	middle_aged	high	yes	fair	yes
13	senior	medium	no	excellent	no

In [14]:
```
# ------------------------------
# Step 2: Separate Features & Label
# ------------------------------
X = dataset.drop("Buys_Computer", axis=1)
y = dataset["Buys_Computer"]
```

In [15]:
```
X
```

Out[15]:

	Age	Income	Student	Credit_Rating
0	youth	high	no	fair
1	youth	high	no	excellent
2	middle_aged	high	no	fair
3	senior	medium	no	fair
4	senior	low	yes	fair
5	senior	low	yes	excellent
6	middle_aged	low	yes	excellent
7	youth	medium	no	fair
8	youth	low	yes	fair
9	senior	medium	yes	fair
10	youth	medium	yes	excellent
11	middle_aged	medium	no	excellent
12	middle_aged	high	yes	fair
13	senior	medium	no	excellent

In [16]:

```
# ------------------------------
# Step 3: Convert Categorical Data to Numeric
# ------------------------------
X = pd.get_dummies(X, drop_first=True)
```

In [17]:

```
# ------------------------------
# Step 4: Train-Test Split
# ------------------------------
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# ------------------------------
# Step 5: Train Logistic Regression
# ------------------------------
#model = LogisticRegression()
model=GaussianNB()
model.fit(X_train, y_train)
```

Out[17]:

```
▾ GaussianNB
GaussianNB()
```

In [18]:

```
# ------------------------------
# Step 6: Evaluate Model
# ------------------------------
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))
```

```
Accuracy: 0.3333333333333333

Classification Report:

              precision    recall  f1-score   support

          no       0.00      0.00      0.00         1
         yes       0.50      0.50      0.50         2

    accuracy                           0.33         3
   macro avg       0.25      0.25      0.25         3
weighted avg       0.33      0.33      0.33         3
```

In [19]:

```
# ------------------------------
# Step 7: Predict Unknown Instance
# Example:
# Age = youth, Income = medium,
# Student = yes, Credit_Rating = fair
# ------------------------------
new_data = pd.DataFrame({
    "Age": ["youth"],
    "Income": ["medium"],
    "Student": ["yes"],
    "Credit_Rating": ["fair"]
})

# Convert to dummy variables
new_data = pd.get_dummies(new_data)

# Match training columns
new_data = new_data.reindex(columns=X.columns, fill_value=0)

prediction = model.predict(new_data)

print("\nPrediction for New Instance:", prediction[0])
```

```
Prediction for New Instance: yes
```

In [2]:

```
import pandas as pd

testdata = {
    "test_size": [0.2, 0.5, 0.6],
    "Accuracy": [0.333, 0.571, 0.555]
}

dataframe = pd.DataFrame(testdata)
dataframe
```

Out[2]:

	test_size	Accuracy
0	0.2	0.333
1	0.5	0.571
2	0.6	0.555

In [3]:

```
rdsdata = {
    "Random State" : [42,50,70],
    "Accuracy" : [0.333,0.666,0.666]
}
```

```
dataframe2 = pd.DataFrame(rdsdata)
dataframe2
```

Out[3]:

	Random State	Accuracy
0	42	0.333
1	50	0.666
2	70	0.666

In [ ]: